

Goursat's Theorem

Theorem 0.0.3. Let G be an open set and let $f : G \rightarrow \mathbb{C}$ be a holomorphic function. Then,

$$\int_T f = 0$$

for every triangular path T in G .

Proof. Let $T_0 = [a, b, c, a]$ be the triangular path in G , and let $d = \text{diam } T$ and $L = \text{length } T$. And, construct four subtriangles

$$T^{(1)} = \left[a, \frac{a+b}{2}, \frac{a+c}{2}, a \right], \quad T^{(2)} = \left[\frac{a+b}{2}, b, \frac{b+c}{2}, \frac{a+b}{2} \right], \quad T^{(3)} = \left[\frac{a+c}{2}, \frac{b+c}{2}, c, \frac{a+c}{2} \right], \quad T^{(4)} = \left[\frac{a+b}{2}, \frac{b+c}{2}, \frac{a+c}{2}, \frac{a+b}{2} \right]$$

Then,

$$\int_{T_0} f = \int_{T_1} f + \int_{T_2} f + \int_{T_3} f + \int_{T_4} f$$

Thus, by the triangle inequality,

$$\left| \int_{T_0} f \right| \leq \left| \int_{T_1} f + \int_{T_2} f + \int_{T_3} f + \int_{T_4} f \right| \leq 4 \cdot \max_{i=1,2,3,4} \left| \int_{T_i} f \right|$$

Let $T_1 = T^j$, where the j maximizes the above integral. Apply the same process for $T_0, T_1, \dots, T_{n-1}$, we obtain that a sequence of nested compact sets $\{\mathcal{T}_n\}$, where $\mathcal{T}_n$ is the convex hull of T_n . So, the intersection

$$\bigcap_{n=0}^{\infty} \mathcal{T}_n = \{z_0\}$$

is a singleton. Given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|z - z_0| < \delta \implies \left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| < \varepsilon$$

since f is holomorphic. Moreover, by the construction, there exists $N \in \mathbb{N}$ such that

$$T_N \subseteq B(z_0; \delta)$$

Now, the triangle inequality again,

$$\left| \int_{T_N} f(z) dz \right| = \left| \int_{T_N} f(z) - f(z_0) - f'(z_0)(z - z_0) dz \right| \leq \varepsilon \int_{T_N} |z - z_0| |dz| = \varepsilon \cdot \text{diam } T_N \cdot \text{length } T_N$$

Finally, since $\text{diam } T_N = 2^{-n} \text{diam } T_0$ and $\text{length } T_N = 2^{-n} \text{length } T_0$,

$$\left| \int_{T_0} f(z) dz \right| \leq 4^N \cdot \left| \int_{T_N} f(z) dz \right| \leq 4^N \cdot \varepsilon \cdot \text{diam } T_N \cdot \text{length } T_N \leq \varepsilon \cdot \text{diam } T_0 \cdot \text{length } T_0 \rightarrow 0 \text{ as } \varepsilon \rightarrow 0$$

□

Thm. Let G be an open set of $\mathbb{C}$, and T be a triangular path contained in G .
 For a point α inside in T , Suppose that f is Holomorphic on $G \setminus \{\alpha\}$.
 If f is bounded on for some neighborhood of α , then

$$\int_T f = 0.$$

Proof. For fixed α , construct a subtriangle T^1, T^2, T^3, T^4 from T .
 Then, for some T^k , $\alpha \in T^k$. (There are at most three cases).

Put this $T_1 = T^k$. Then, inductively, we can construct $\{T_n\}$, compact nested, and

$$\left| \int_T f \right| \leq \left| \sum_{k \in T_n^k} \int_{T_n^k} f \right| \leq \frac{3^n}{2^n} \cdot \sup |f| \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Morera Theorem.

Let $f: D \rightarrow \mathbb{C}$ be a continuous function on an open set D .
 Suppose that for every triangle T inside G ,

$$\int_T f = 0.$$

Then, f is Holomorphic on G .

Proof. Let $z_0 \in G$ be fixed, and $B_r(z_0)$ be given where r was chosen by $B_r(z_0) \subseteq G$. For each $z \in B_r(z_0)$, define

$$F(z) = \int_{\gamma_{z_0, z}} f(u) du \text{ where } \gamma_{z_0, z} \text{ is the straight segment.}$$

For small $h \in \mathbb{C}$, observe that

$$\frac{F(z+h) - F(z)}{h} = \frac{1}{h} \int_{\gamma_{z, z+h}} f(u) du, \text{ because}$$

$$\int_{\gamma_{z_0, z}} f + \int_{\gamma_{z, z+h}} f + \int_{\gamma_{z+h, z_0}} f = 0, \text{ by the assumption.}$$

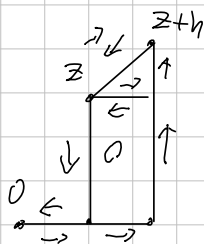
Now, the continuity gives $\lim_{h \rightarrow 0} \int_{\gamma_{z, z+h}} f(u) du = f(z) \cdot h$. $\square$



Thm. A holomorphic function $f: D \rightarrow \mathbb{C}$ has a primitive where D is open disk.

Proof. WLOG, let $D = B(0; R)$. Given $z = a + bi \in D$, define

$$F(z) = \int_{\gamma_z} f(s) ds \quad \text{where } \gamma_z = [(0,0), (a,0), (a,b)]$$



Now, let $h \in \mathbb{C}$ with $0 < |h| < R - |z|$. Then,

$$F(z+h) - F(z) = \int_{\gamma_{z+h}} f(s) ds - \int_{\gamma_z} f(s) ds = \int_{\gamma_{z,z+h}} f(s) ds$$

using Gauss's Thm in Here, with holomorphic.

Given $\epsilon > 0$, there is $\delta > 0$ such that

$$|z - s| < \delta \Rightarrow |f(z) - f(s)| < \epsilon.$$

So,
$$\left| \frac{1}{h} \cdot \int_{\gamma_{z,z+h}} f(s) ds - f(z) \right| = \left| \frac{1}{h} \cdot \int_{\gamma_{z,z+h}} f(s) - f(z) ds \right| < \epsilon, \text{ for small enough } h.$$

Corollary. If f is holomorphic on an open disk, then for any closed curve γ in the disk,

$$\int_{\gamma} f = 0.$$

Cauchy integral formula

Thm. Let $D = \{z \in \mathbb{C} \mid |z - z_0| \leq r\}$ be a disk centered at z_0 , and $C: t \mapsto z_0 + r \cdot e^{it}$ be a curve, along the boundary of D .

Suppose that $f: G \rightarrow \mathbb{C}$ is a holomorphic on an open G containing D .

Given $z \in D$,
$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(u)}{u - z} du$$

Proof. Let $z \in D$ be fixed. Since $f(u)/(u - z)$ is holomorphic except $u = z$, for a curve $C_\epsilon: t \mapsto z + \epsilon \cdot e^{it}$ where $\epsilon < r$, we obtain

$$\int_{-C_\epsilon} \frac{f(u)}{u - z} du + \int_C \frac{f(u)}{u - z} du = 0, \quad \dots (*)$$

by the Cauchy-Goursat theorem.

Since f is holomorphic, the derivative of f near z is bounded, so

$$\int_{C_\epsilon} \frac{f(u) - f(z)}{u - z} du \rightarrow 0 \quad \text{as } \epsilon \rightarrow 0.$$

and, for any $\epsilon > 0$,

$$\int_{C_\epsilon} \frac{f(z)}{u - z} du = f(z) \cdot \int_0^{2\pi} \frac{1}{\epsilon \cdot e^{it}} \cdot i \cdot \epsilon e^{it} dt = f(z) \cdot 2\pi i.$$

Therefore,

$$\int_{-C_\epsilon} \frac{f(u)}{u - z} du = - \int_{C_\epsilon} \frac{f(u) - f(z)}{u - z} du - \int_{C_\epsilon} \frac{f(z)}{u - z} du \rightarrow -2\pi i \cdot f(z),$$

as $\epsilon \rightarrow 0$, so we obtain the result, taking $\epsilon \rightarrow 0$ to $(*)$. $\square$

Thm. Suppose f is Holomorphic on an open G containing a closed disk $\overline{D}$ with center z_0 ,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n \quad (z \in D)$$

Proof. $f(z) = \frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{\mu - z} d\mu \quad \left(\frac{1}{\mu - z} = \frac{1}{\mu - z_0 - (z - z_0)} = \frac{1}{\mu - z_0} \cdot \frac{1}{1 - \frac{z - z_0}{\mu - z_0}} \right)$

$$= \frac{1}{2\pi i} \cdot \int_C f(\mu) \cdot \sum_{n=0}^{\infty} \frac{1}{(\mu - z_0)^{n+1}} \cdot (z - z_0)^n d\mu$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{(\mu - z_0)^{n+1}} d\mu \right) \cdot (z - z_0)^n$$

$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n$$

Corollary, Liouville's Theorem.

If $f: \mathbb{C} \rightarrow \mathbb{C}$ is entire and bounded, then it is constant.

Proof. Given $z_0 \in \mathbb{C}$ and $R > 0$, $|f(z_0)| \leq \frac{1}{R} \cdot \sup_{\mathbb{C}_R} |f| \rightarrow 0$ as $R \rightarrow \infty$. Therefore, $f' = 0$.
 Thus f is constant.

Fundamental Theorem of Algebra.

Non-constant polynomial $P(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_0 \in \mathbb{C}[z]$ has a root in $\mathbb{C}$.

Proof. Suppose that $P(z)$ has no root in $\mathbb{C}$. Then, $1/P(z)$ is well-defined on $\mathbb{C}$, entire. observe that
$$\frac{P(z)}{z^n} = a_n + \left(\frac{a_{n-1}}{z} + \dots + \frac{a_0}{z^n} \right) \quad (z \neq 0).$$

The terms $a_{n-k} \cdot z^{-k}$ ($k=1, \dots, n$) converges to zero as $|z| \rightarrow \infty$, thus for some $R > 0$,

$$|P(z)| \geq 2|a_n| \cdot |z|^n \quad (|z| > R)$$

$$\geq 2|a_n| \cdot R^n$$

Therefore it is bounded below in $|z| > R$, and clearly in $|z| \leq R$.

Now, $\frac{1}{P}$ is bounded, thus constant. Contradiction.